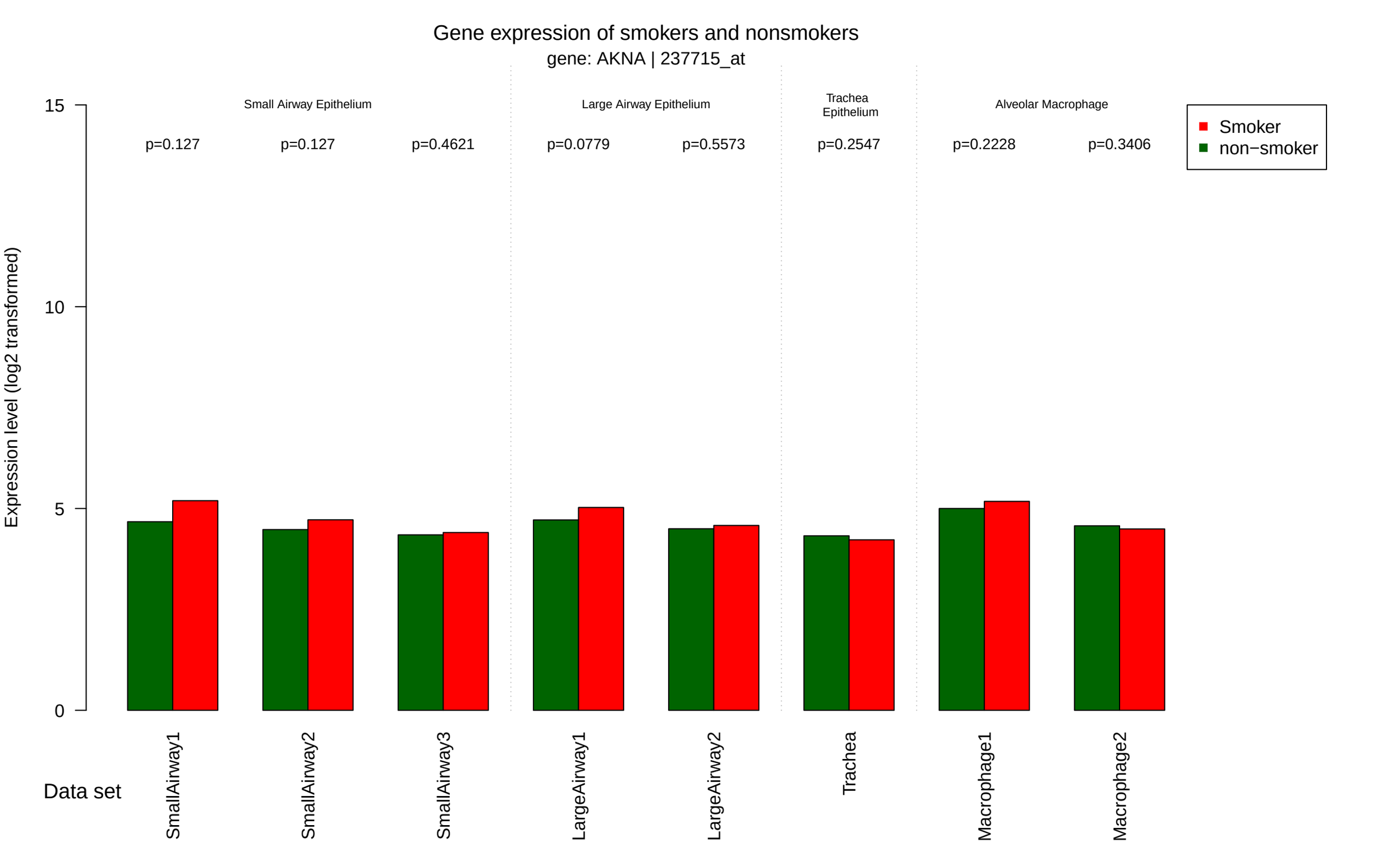
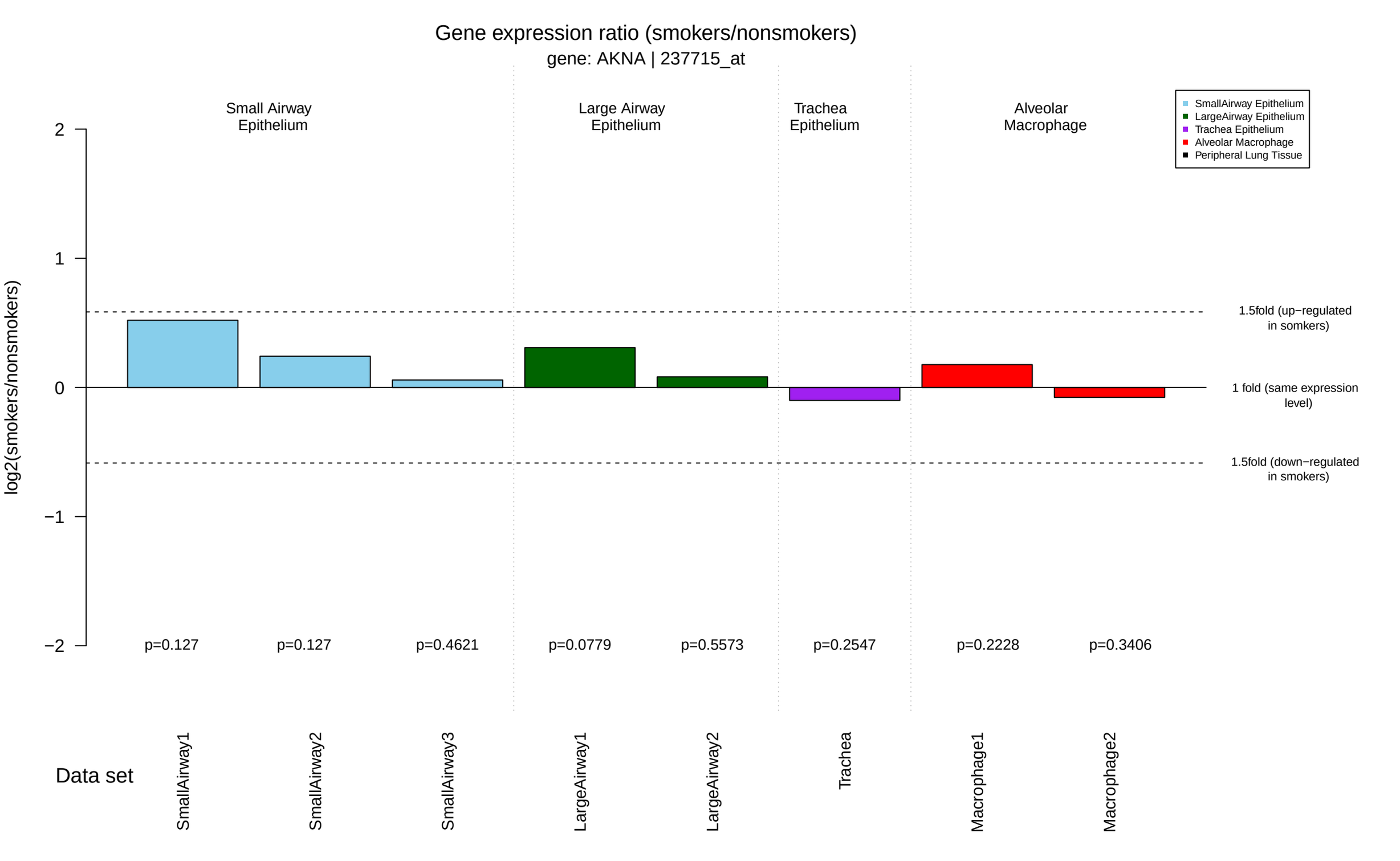
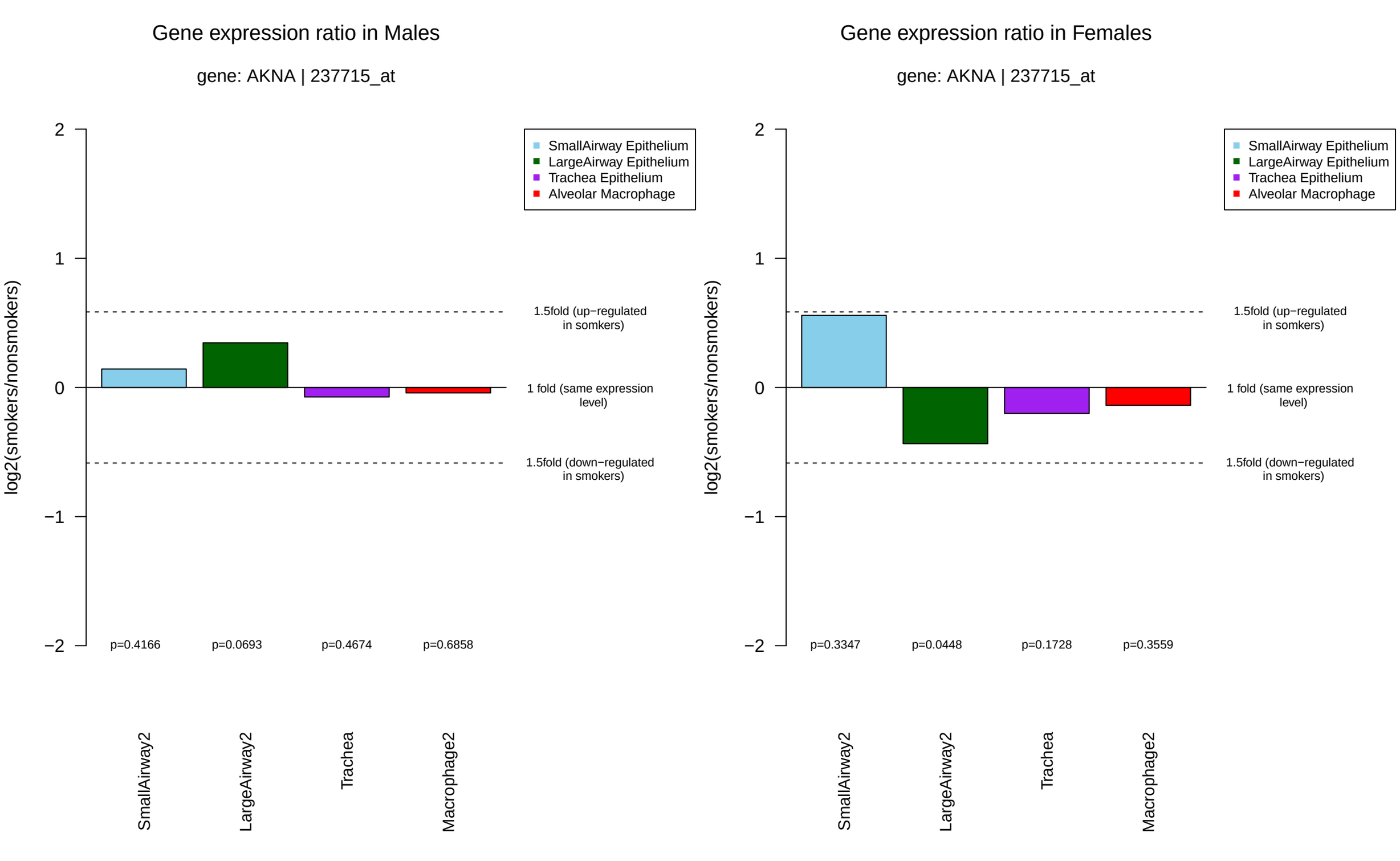
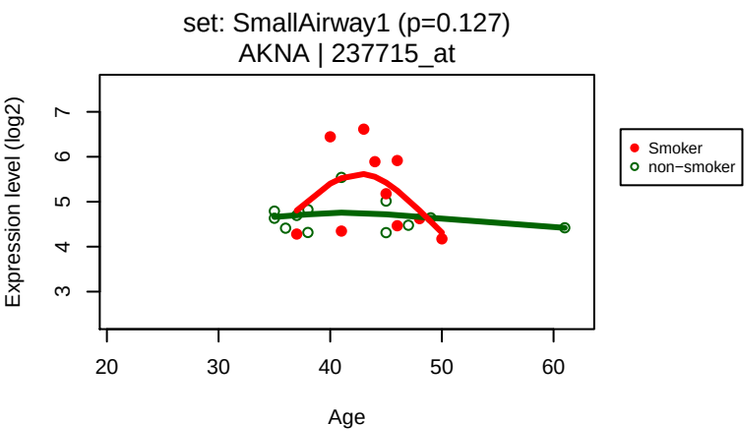
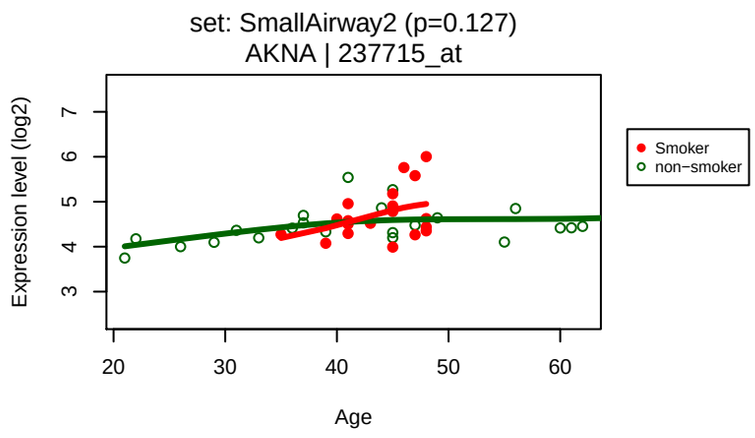
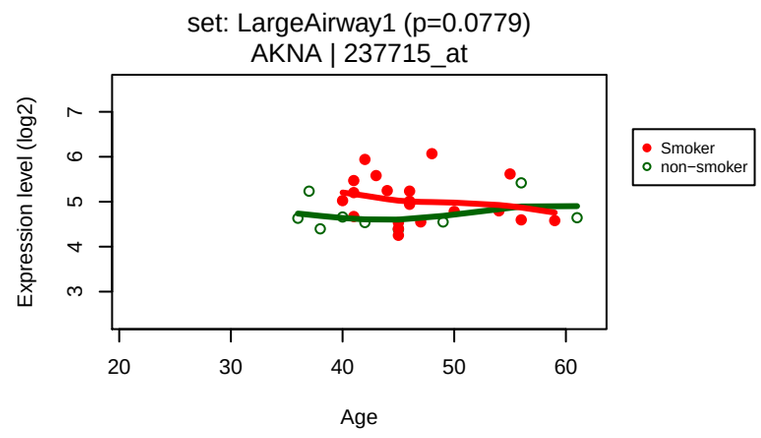

Gene expression of smokers and nonsmokers
gene: AKNA | 237715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.127
p=0.127
p=0.4621
p=0.0779
p=0.5573
p=0.2547
p=0.2228
p=0.3406
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

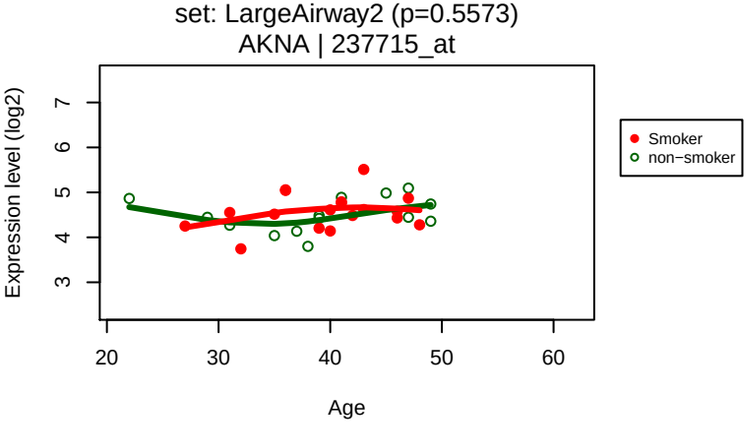

Gene expression ratio (smokers/nonsmokers)
gene: AKNA | 237715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.127
p=0.127
p=0.4621
p=0.0779
p=0.5573
p=0.2547
p=0.2228
p=0.3406
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

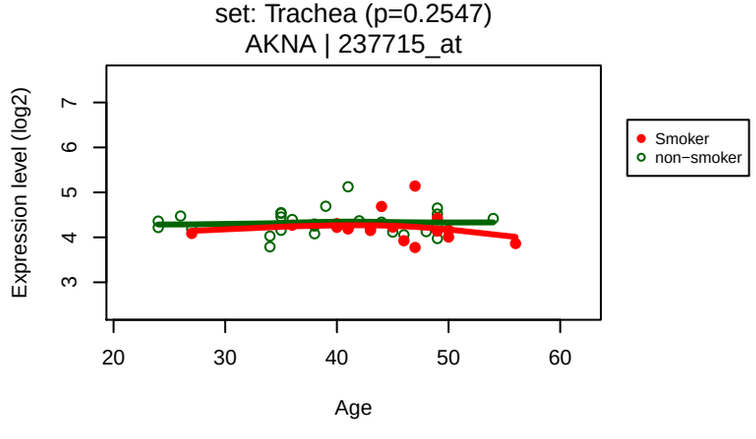

Gene expression ratio in Males
gene: AKNA | 237715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4166
p=0.0693
p=0.4674
p=0.6858
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKNA | 237715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3347
p=0.0448
p=0.1728
p=0.3559
SmallAirway2
LargeAirway2
Trachea
Macrophage2

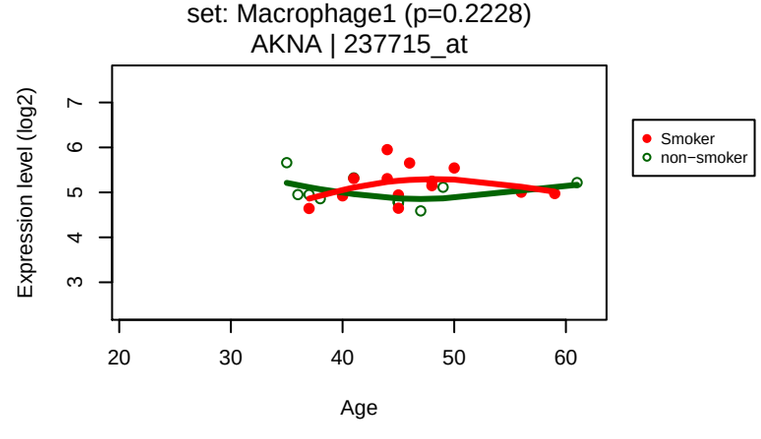

set: SmallAirway1 (p=0.127)
AKNA | 237715_at
Expression level (log2)
Age
Smoker
non−smoker

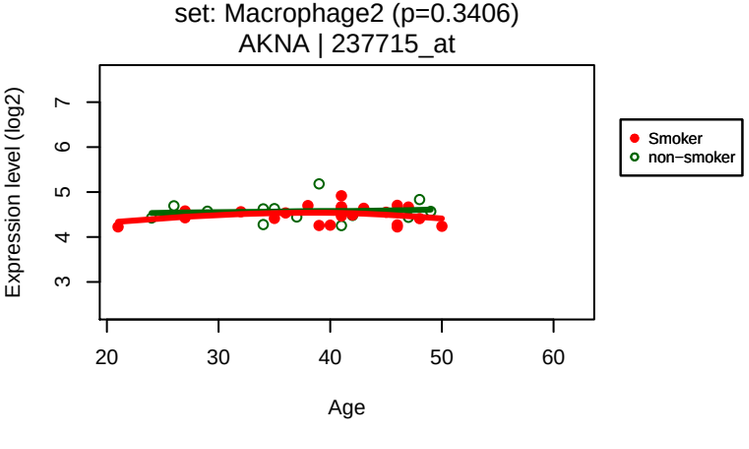

set: SmallAirway2 (p=0.127)
AKNA | 237715_at
Expression level (log2)
Age
Smoker
non−smoker

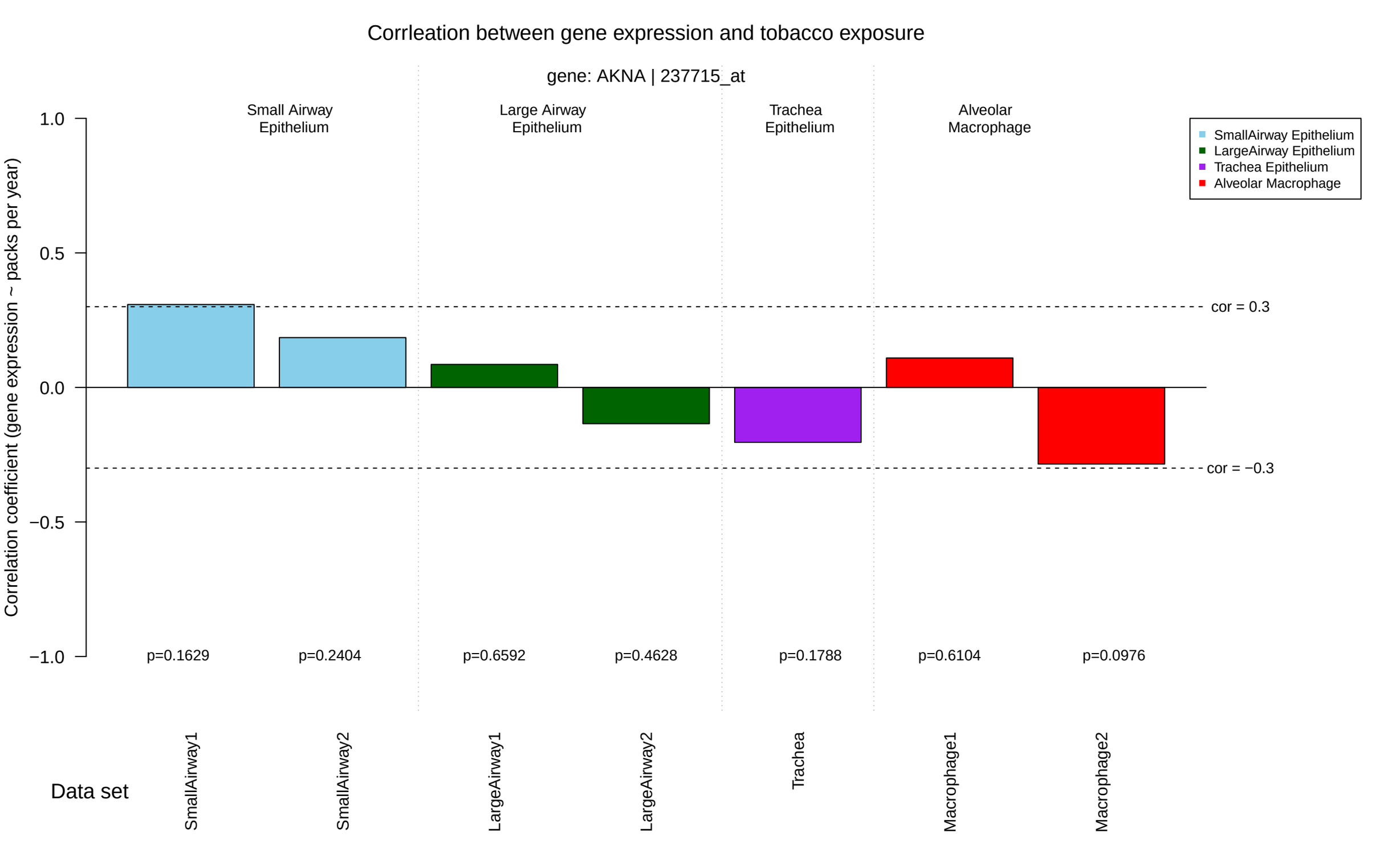

set: LargeAirway1 (p=0.0779)
AKNA | 237715_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5573)
AKNA | 237715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2547)
AKNA | 237715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2228)
AKNA | 237715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3406)
AKNA | 237715_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKNA | 237715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1629
p=0.2404
p=0.6592
p=0.4628
p=0.1788
p=0.6104
p=0.0976
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2